

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 50

Annexure A

MCQ

Code of Conduct:

I K PHANI SRIDHAR YOGI/ 192311091 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Annexure A

MCQ

MCQ (Additional Set)

1. Which property ensures the same input always produces the same hash?
 - A) Avalanche effect
 - B) Determinism
 - C) Collision resistance
 - D) Diffusion
2. What is the primary purpose of SHA-3?
 - A) Faster encryption
 - B) Alternative secure hashing standard
 - C) Data compression
 - D) Key exchange
3. Which attack tries all possible inputs to match a hash?
 - A) Replay attack
 - B) Brute-force attack
 - C) MITM attack
 - D) DoS attack
4. What is a salted hash?
 - A) Encrypted hash
 - B) Hash with random data added
 - C) Shortened hash
 - D) Reversed hash
5. Which algorithm is widely used for password hashing?
 - A) AES
 - B) RSA
 - C) bcrypt
 - D) DES
6. What does second pre-image resistance ensure?
 - A) Same input produces same output
 - B) Hard to find different input with same hash

- C) Fast hashing
- D) Small hash size

7. What is the output size of SHA-512?

- A) 256 bits
- B) 128 bits
- C) 512 bits
- D) 1024 bits

8. Which technique slows down brute-force attacks?

- A) Encryption
- B) Key exchange
- C) Key stretching
- D) Compression

9. What is the purpose of a nonce?

- A) Encryption
- B) Prevent replay attacks
- C) Compress data
- D) Generate keys

10. Which function is irreversible?

- A) Encryption
- B) Hashing
- C) Compression
- D) Encoding

11\ . Which algorithm is used in DSS?

- A) AES
- B) DSA
- C) DES
- D) RC4

12\ . What is required to verify a digital signature?

- A) Private key
- B) Public key
- C) Session key
- D) Shared key

13\ What does a digital signature guarantee?

- A) Confidentiality
- B) Integrity and authenticity
- C) Compression
- D) Speed

14\ What happens if message digest changes?

- A) Signature remains valid
- B) Verification fails
- C) Encryption fails
- D) Speed increases

15\ Which attack involves forging a signature?

- A) Replay
- B) Forgery
- C) DoS
- D) Sniffing

16\ What is the role of a hash in signatures?

- A) Encrypt data
- B) Reduce size
- C) Ensure integrity
- D) Store keys

17\ Which key is kept secret in signatures?

- A) Public key
- B) Private key
- C) Session key
- D) Shared key

18\ What ensures trust in certificates?

- A) Encryption
- B) Certificate Authority
- C) Compression
- D) Hash size

19\ What is certificate revocation?

- A) Key generation

B) Canceling validity

C) Encryption

D) Compression

20\.. What is OCSP used for?

A) Encryption

B) Certificate validation

C) Compression

D) Routing

21\.. What is the main goal of Kerberos tickets?

A) Encryption

B) Authentication

C) Compression

D) Routing

22\.. Which protocol uses tickets for authentication?

A) SSL

B) Kerberos

C) FTP

D) HTTP

23\.. What is a session key?

A) Permanent key

B) Temporary key

C) Public key

D) Private key

24\.. What is mutual authentication?

A) One-way authentication

B) Both parties verify each other

C) Anonymous login

D) Weak authentication

25\.. What ensures freshness in Kerberos?

A) Encryption

B) Timestamp

C) Compression

D) Hashing

26\.. What is a replay attack?

A) Data modification

B) Resending valid data

C) Encryption

D) Compression

27\.. What is the function of an authenticator in Kerberos?

A) Data storage

B) Identity proof

C) Encryption

D) Compression

28\.. What happens if system clocks differ in Kerberos?

A) Faster authentication

B) Failure

C) Better security

D) No effect

29\.. What is the KDC composed of?

A) Client & Server

B) AS & TGS

C) Firewall & IDS

D) Router & Switch

30\.. What is the AS in Kerberos?

A) Application Server

B) Authentication Server

C) Access Server

D) Admin Server

31\.. What is X.509 certificate format used in?

A) Networking

B) Digital certificates

C) Compression

D) Routing

32\.. What identifies the certificate owner?

- A) Public key
- B) Subject field

- C) Signature
- D) Hash

33\.. What ensures certificate integrity?

- A) Encryption
- B) Digital signature
- C) Compression
- D) Routing

34\.. What is a root certificate?

- A) Expired certificate
- B) Self-signed trusted certificate
- C) Invalid certificate
- D) Temporary certificate

35\.. What is certificate pinning?

- A) Storing keys
- B) Binding certificate to host
- C) Encryption
- D) Compression

36\.. What is the risk of weak CA validation?

- A) Faster speed
- B) MITM attacks
- C) Compression
- D) Routing

37\.. What is the purpose of trust chain?

- A) Encryption
- B) Validate certificate hierarchy
- C) Compress data
- D) Routing

38\.. What is digital certificate used for?

- A) Compression

B) Authentication

C) Routing

D) Storage

39\.. What ensures non-repudiation?

A) Encryption

B) Digital signature

C) Compression

D) Routing

40\.. What is a hash collision?

A) Same input

B) Same output for different inputs

C) Encryption

D) Compression

41\.. Which hash is strongest?

A) MD5

B) SHA-1

C) SHA-256

D) CRC

42\.. What is a one-way function?

A) Reversible

B) Irreversible

C) Fast

D) Small

43\.. What ensures authentication in PKI?

A) Encryption

B) Certificates

C) Compression

D) Routing

44\.. What is key escrow?

A) Key deletion

B) Key storage by third party

C) Encryption

D) Compression

45\.. What is identity spoofing?

A) Encryption

B) Fake identity

C) Compression

D) Routing

46\.. What is trust anchor?

A) Root CA

B) Client

C) Server

D) Router

47\.. What ensures integrity of message?

A) Encryption

B) Hash

C) Compression

D) Routing

48\.. What is digital envelope?

A) Hash

B) Encrypted message + key

C) Compression

D) Routing

49\.. What is certificate fingerprint?

A) Encryption

B) Hash of certificate

C) Compression

D) Routing

50\.. What is PKI used for?

A) Routing

B) Secure communication

C) Compression

D) Storage

Q.No	Answer	Explanation
1	B) Determinism	Same input always gives the same hash output.
2	B) Alternative secure hashing standard	SHA-3 is a secure alternative to SHA-2.
3	B) Brute-force attack	Tries every possible input until a match is found.
4	B) Hash with random data added	Salt is random data added before hashing.
5	C) bcrypt	bcrypt is widely used for password hashing.
6	B) Hard to find different input with same hash	Second pre-image resistance prevents finding another input with the same hash as a given input.
7	C) 512 bits	SHA-512 produces a 512-bit hash.
8	C) Key stretching	Makes brute-force attacks much slower.
9	B) Prevent replay attacks	Nonces ensure freshness.
10	B) Hashing	Hash functions are one-way (irreversible).
11	B) DSA	DSS uses the Digital Signature Algorithm (DSA).
12	B) Public key	Digital signatures are verified using the sender's public key.
13	B) Integrity and authenticity	Digital signatures verify origin and integrity.
14	B) Verification fails	Any message change changes the digest.
15	B) Forgery	Forgery attempts to create fake signatures.
16	C) Ensure integrity	Hashes detect message modifications.
17	B) Private key	The signer keeps the private key secret.
18	B) Certificate Authority	Trusted CAs issue certificates.
19	B) Canceling validity	Revocation invalidates a certificate before expiry.
20	B) Certificate validation	OCSP checks certificate status.
21	B) Authentication	Kerberos tickets authenticate users securely.
22	B) Kerberos	Kerberos uses tickets for authentication.
23	B) Temporary key	Session keys are temporary.
24	B) Both parties verify each other	This is mutual authentication.
25	B) Timestamp	Timestamps prevent replay attacks.

Q.No	Answer	Explanation
26	B) Resending valid data	Replay attacks resend captured valid messages.
27	B) Identity proof	Authenticator proves client identity.
28	B) Failure	Kerberos relies on synchronized clocks.
29	B) AS & TGS	KDC contains Authentication Server and Ticket Granting Server.
30	B) Authentication Server	AS authenticates users initially.
31	B) Digital certificates	X.509 is the standard certificate format.
32	B) Subject field	Subject identifies the certificate owner.
33	B) Digital signature	CA's digital signature protects certificate integrity.
34	B) Self-signed trusted certificate	Root certificates are self-signed.
35	B) Binding certificate to host	Certificate pinning prevents fake certificates.
36	B) MITM attacks	Weak validation enables Man-in-the-Middle attacks.
37	B) Validate certificate hierarchy	Trust chain verifies certificates up to the Root CA.
38	B) Authentication	Certificates authenticate identities.
39	B) Digital signature	Digital signatures provide non-repudiation.
40	B) Same output for different inputs	This is a hash collision.
41	C) SHA-256	SHA-256 is stronger than MD5 and SHA-1.
42	B) Irreversible	One-way functions cannot be reversed efficiently.
43	B) Certificates	PKI authenticates users using certificates.
44	B) Key storage by third party	Key escrow stores keys with a trusted third party.
45	B) Fake identity	Identity spoofing means pretending to be another entity.
46	A) Root CA	Root CA serves as the trust anchor.
47	B) Hash	Hashing verifies message integrity.
48	B) Encrypted message + key	A digital envelope combines encrypted data and an encrypted session key.
49	B) Hash of certificate	Fingerprint uniquely identifies a certificate.
50	B) Secure communication	PKI enables secure communication through certificates and keys.

